

Rajdeep Pal

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Professional Summary

Cybersecurity and AI-focused B.Tech CSE student with hands-on experience in threat detection engineering, SecOps automation, and cognitive security intelligence. Proven track record in building offline-first threat simulation platforms, orchestrating containerized microservices, and developing active NIDS evasion frameworks. Skilled in Wazuh, Suricata, Docker Compose, and local-AI integration to optimize enterprise security posture.

Education

B.Tech CSE (Cybersecurity & Forensics) – UPES, Dehradun	2023 – Present CGPA: 7.75
Class XII (CBSE) – Amity International School, Gurgaon	2022 – 2023 84.2%
Class X (CBSE) – Amity International School, Gurgaon	2020 – 2021 91.7%

Technical Skills

- **Programming & Scripting:** Python (FastAPI, SQLAlchemy, Celery, Paramiko), JavaScript (Vanilla JS, WebSockets), HTML5/CSS3, SQL (PostgreSQL), Java, C++
- **Infrastructure & DevOps:** Vulnerability Scanners, Kubernetes (Minikube), Docker, Red Hat Enterprise Linux 9 (RHEL 9), Git, Bash, PowerShell, Postman, Tkinter, VirtualBox, Kali Linux, Ubuntu, Windows
- **AI-Security Integration:** LLM API Integration (Gemini 2.5 Flash, Claude), Autonomous ReAct Agent Loops, Cognitive Security Planning, Automated Remediation Code Generation
- **Security Concepts:** OWASP Top 10, Static/Dynamic Application Security Testing (SAST/DAST), Cryptographic Guardrails, Network Intrusion Detection Systems (NIDS) Evasion, Threat Modeling

Experience

Cybersecurity Intern (Security Product Testing and Penetration)	June 2026 – Present
Jio Platforms Limited, Navi Mumbai	

- **Jio Cybersecurity Suite (JCS):** Engineered a unified, offline-first cognitive security orchestration platform combining JCS0 (automated threat intelligence & CVE scanner) and JCS1 (autonomous pentesting & threat simulation agent).
- **Subsystem Capabilities:** Developed JCS0 to automate passive/active target reconnaissance with offline SQLite CVE correlation, and built JCS1 as an interactive ReAct reasoning agent executing active auditing modules (including credential audits, NIDS evasion, and DDoS simulations) behind manual operator safety gates.
- **System Architecture:** Deployed the decoupled pipeline (FastAPI, PostgreSQL, Redis, and three segregated Celery worker tiers) via Docker Compose on RHEL 9, securing external access through a host-networked Nginx reverse proxy.

Cybersecurity Intern

Jun 2025 – Jul 2025

Power Grid Corporation of India Limited, Gurgaon

- Developed and deployed two internal Python log analysis tools, reducing manual SOC triage effort by approximately 40%.
- Performed security testing and system behavior simulations to identify vulnerabilities, collaborating with SOC analysts to understand SIEM pipelines, alert prioritization, and threat intelligence integration.

Projects

Jio Cybersecurity Suite (JCS)	Python, FastAPI, Docker Compose, Ollama
Jio Platforms Limited (Internal Security Tooling)	

- **JCS0 & JCS1 Subsystems:** Developed JCS0 (automated passive/active DNS and TCP port sweeps with offline SQLite CVE correlation) and JCS1 (autonomous pentesting agent executing 10 active auditing modules including NIDS evasion and protocol desynchronization).
- **Local-AI & Orchestration:** Deployed the FastAPI/PostgreSQL/Redis microservices pipeline via Docker Compose on RHEL 9, utilizing a local, self-hosted LLM (Ollama with phi3) via a custom compatibility adapter (OllamaClientCompat) for 100% offline data privacy.

TrackrAI Job Tracker

React, FastAPI, JWT Auth, SQLite

github.com/rajdeppal01/trackrai

- **Multi-User Authentication:** Engineered a JWT-based authentication system with bcrypt password hashing, enabling per-user data scoping and complete data isolation across a shared FastAPI backend.
- **Full-Stack Job Pipeline:** Built a React/Vite frontend featuring a drag-and-drop Kanban pipeline (Applied to Offer), a live analytics dashboard, and an AI Copilot module that scores pipeline health and generates personalized next-step recommendations.

Portfolio

Vanilla JS, HTML5/CSS3

rajdeeppal01.github.io/portfolio

- Designed and built a fully interactive terminal-emulator portfolio from scratch in vanilla JavaScript, where every section is navigable via live shell commands (`cd`, `ls`) rather than a conventional scrolling layout.
- Implemented a hover-expandable showcase gallery with lazy-loaded iframe previews of live deployed projects, plus multiple custom color themes and dynamic section rendering, deployed via GitHub Pages.

Event Viewer GUI Tool

PowerShell, WinForms

github.com/rajdeeppal01/custom-event-viewer

- Engineered a lightweight WinForms-based alternative to Windows Event Viewer for SOC log triage, reducing manual analysis time by approximately 30% through advanced custom rule filtering.

Windows Event Log Parser

Python, Tkinter

github.com/rajdeeppal01/custom-log-parser

- Developed a cross-version Python/Tkinter log parser for Windows and Apache web server logs, implementing keyword, severity, and time-window anomaly detection to identify brute-force attacks.

Certifications

AWS Security Learning Plan (2025) • Deloitte, TATA, and Mastercard Cyber Job Simulations – Forage (2025) • ISRO: AI/ML for Geodata Analysis (2025) • ISC2 Certified in Cybersecurity (CC) (2025)

Achievements

IEEE CIS Student Chapter – PR Associate Head (2025 – 2026)